Supplementary Material

Cross-Modal Safety-Component Steering for Audio-Language Model Jailbreak Defense

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This appendix provides extended protocols and experimental analyses for Cross-Modal Safety-Component Steering, an inference-time defense that applies calibrated safety directions to selected decoder heads and native audio-interface components. The main study evaluates Qwen2-Audio-7B-Instruct and MOSS-Audio-8B-Instruct across six non-adaptive attack families and three benign listening tasks. The detailed component, comparator, window, adaptive-attack, and resource analyses in this appendix focus on Qwen2-Audio.

Local steering updates selected states during prefill and the first eight generation steps. Conditional steering activates decoder interventions through a calibrated risk gate; family-routed steering additionally selects a family-specific direction. Both gated policies reuse an unmodified prefill cache. All three policies preserve model weights, prompts, and decoding settings.

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Table 1: Fixed-signal validation-weight sensitivity analysis, including the calibrated default configuration.

Configuration	Shared/5	Jaccard	Unsafe/180	BUP	FPR
Default (calibrated)	5	1	19 (10.6%)	96.2%	7.2%
Equal absolute coefficients	4	0.67	20 (11.1%)	96.3%	6.7%
Half benign penalty	3	0.43	18 (10%)	95.4%	8.9%†
Double benign penalty	4	0.67	22 (12.2%)	97%	5.6%
Half intervention-evidence group	4	0.67	21 (11.7%)	96.5%	6.7%
Double intervention-evidence group	4	0.67	18 (10%)	95.8%	8.3%†

$K = 5$; 180 attacks and 180 benign items. False-refusal counts are $13, 12, 16, 10, 12, 15$. † denotes FPR above 8% at the fixed local strength (more than 14 false refusals), not statistical significance or infeasibility after recalibration. Equal absolute coefficients are $(1, 1, 1, 1, -1, 1)$. Default bootstrap frequencies remain fixed.

1 Definitions and Controlled Direction Estimation

Quantity	Meaning
$h \in \mathcal{C}$	Decoder head or architecture-native audio-interface component.
$\bar{\mathbf{v}}_h$	Unit-normalized, component-specific intervention direction.
ζ / $\alpha^\star$	Local diagnostic perturbation / frozen deployment strength.
G_h	Signed first-order change in the safety-margin surrogate.
B_h	Expected absolute first-order change in the benign reference-response score.
$r(x), \tau^\star$	Prefill risk score and frozen gate threshold.
ASR / BUP / FPR	Unsafe-assistance rate / average task-score retention / unnecessary refusal, verification, or deferral rate.

The absolute value in $B_h = \mathbb{E}[|\partial u_h / \partial \zeta|_{\zeta=0}]$ makes it a sensitivity magnitude. For example, derivatives $+0.02$ and -0.02 both have magnitude 0.02. A negative coefficient penalizes reference-score disturbance in either direction. This local surrogate is distinct from decoded task utility: BUP evaluates task-score retention, while FPR evaluates unnecessary refusal, verification, or deferral. Consequently, BUP need not equal one minus FPR.

Teacher forcing controls the reference prefix and response positions used for matched direction estimation. Autoregressive deployment conditions on the generated prefix, so the state distributions can differ. The behavioral evaluation tests whether the extracted directions are useful during actual decoding; it does not require or establish identical states. This distinction provides a controlled estimator together with a deployment-level test, while leaving prefix divergence as a limitation beyond the evaluated operating window.

2 Ranking Weights and the Short Intervention Window

The coefficient order is ablation, patching, margin gain, audio evidence, benign sensitivity, and bootstrap stability. The default $(2, 1.5, 2, 1, -2, 1)$ gives greater emphasis to intervention effects, downweights patching for alignment uncertainty, penalizes benign sensitivity, and retains audio evidence and stability as auxiliary signals.

The coefficient analysis is a *fixed-signal sensitivity analysis*. The candidate inventory, discovery normalization, direction estimates, and default bootstrap-frequency signal are held fixed while coefficients are perturbed. Component sets retain $K = 5$ and use the calibrated local strength and eight-token window. This specification isolates how reweighting the stored signals affects selection and validation behavior; it does not retrain the discovery procedure or recompute bootstrap frequencies for each stability-free score. Validation uses 180 attack and 180 benign items, separately from the test comparisons in Section 3.

Equal absolute coefficients give 20/180 unsafe responses, 96.3% BUP, and 6.7% FPR, versus 19/180, 96.2%, and 7.2% for the default. Halving the benign penalty lowers ASR but raises FPR to 8.9%; doubling it gives 97% BUP with 12.2% ASR. The daggered configurations exceed the validation FPR cap only at the frozen strength and may become feasible after recalibration. For $K = 5$ and overlap m, the reported Jaccard index is $m/(10 - m)$; replacing two sites changes 40% of the selected set. Because nearby aggregate rates can conceal different component sets, we report selection overlap separately and do not claim a universally optimal coefficient vector or a monotone sensitivity–FPR relation.

The configuration search considers 4-, 8-, and 16-token windows. The eight-token setting limits the duration and cost of the update, rather than the duration of audio reasoning. After the intervention window, audio-conditioned decoding continues without further steering updates. Early intervention can influence the subsequent trajectory, while the acoustic pathway remains part of the model. The AIR-Bench, Clotho, and AudioSet results measure retention on their stated tasks; sufficiency for arbitrary long-form, multi-step, or streaming generation is outside this evidence.

3 Related Methods and Targeted Controls

CrossSteer is described at abstract level as learning a text-only preference direction in the shared residual stream at an audio-robust layer [1]. Only the title and abstract were accessible for this comparison; we therefore restrict our discussion to that description and do not infer unavailable implementation details.

SARSteer retains audio while contrasting refusal-text conditions within the same LALM and removes a benign PCA subspace [3]. ITI provides a sparse head-intervention precedent [2]; CAA provides contrastive residual addition and layer selection [4]. Our contribution concerns component-specific audio-conditioned directions, heterogeneous native sites, intervention-based selection, and calibrated deployment. The internal controls isolate parts of that construction. A separate paper-based SARSteer adaptation is evaluated below; it is distinct from an upstream implementation reproduction. No CrossSteer or ITI performance is assigned.

The comparison protocol has two complementary internal controls and one close-method adaptation. The *layer-calibrated residual control* fits a matched mean-contrast direction at each candidate residual site using extraction data, then selects layer and strength using the common constrained validation objective. The protocol considers decoder layers $8, 16, 24, 32$ (one-based, including the final decoder layer) and strengths $\{1, 1.4, 1.8, 2.2, 2.6, 3, 3.4\}$. The *safety-gain-ranked component control* retains the same heterogeneous candidate pool, whitened component-specific directions, $K = 5$, support, and generation window, while replacing the composite rank by G_h alone and selecting its own strength from that grid under the same constraints. Rank ties use a fixed component identifier. Direction fitting uses extraction data, ranking uses discovery data, and configuration choices use validation data. Feasible configurations are selected and frozen before the common 760-attack/300-benign test evaluation, with the same checkpoint, preprocessing, decoding, and response judging. The component budget is matched for the ranking control; this does not assert identical tuning costs across different method families.

The *SARSteer protocol adaptation* follows the paper's refusal-text contrast and safe-space ablation, rather than relabelling our independent text-model baseline. For each harmful extraction input, the audio is unchanged and the text instruction is evaluated with and without the appended refusal anchor "I cannot assist with that." Last-input-token residual contrasts yield one direction per layer. Centered benign activations define the PCA subspace; its projection is removed without converting the remaining vector to the proposed method's unit-norm convention. All 32 decoder layers, PCA ranks 5/10/20, and native strengths 0.025/0.05/0.1/0.2/0.4 are eligible. This grid includes the paper's $k = 10$, $\alpha = 0.1$ setting; layer, rank and strength are selected only on

Table 2: Common-test comparison of local steering with calibrated residual and component-ranking controls and a paper-based SARSteer adaptation. No paired confidence intervals or significance tests are reported.

Method	Unsafe/760	ASR	BUP	FPR	Overhead (ms)
Local steering	79	10.4%	96.1%	7.3%	1.6
Final-layer residual	92	12.1%	90.1%	15.7%	1.2
Layer-calibrated residual control	84	11.1%	95.7%	8%	1.3
Safety-gain-ranked component control	88	11.6%	95.2%	9%	1.6
SARSteer (protocol adaptation)	82	10.8%	96.4%	6.3%	1.4

FPR uses 300 benign items: local/final-layer residual have 22/47 false refusals; calibrated residual/gain-only/SARSteer have counts 24/27/19. BUP is task-score retention. The 8% FPR rule applies to validation. Calibrated residual changes layer, strength, and operating-point selection. SARSteer is a paper-based extra-data protocol adaptation with method-specific fitting and calibration, distinct from an upstream implementation reproduction.

the common constrained validation objective. Updates run at every generated token up to the unchanged 256-token cap, rather than imposing our eight-token window. The refusal anchor is used for fitting, while the deployment prompt is unchanged.

The adaptation uses 120 additional benign fitting clips for PCA, group-disjoint from all primary partitions. The fitting protocol excludes shared source clips, speakers or voices, backgrounds, room responses, device profiles, and goal or prompt groups wherever those factors apply. The fitting record comprises the PCA sample lists, centering, singular values, selected rank, and searched configurations; no PCA-stability measurement is reported. This extra-data variant is a method-specific fitting-budget comparison. The comparison shares checkpoint, test items, scoring, decoding, and selection constraints, but does not isolate method differences at identical fitting-data or tuning budgets. It uses the paper-based adaptation rather than the upstream implementation, and does not transfer the external paper's published benchmark scores into our test suite.

Task-level retention for these methods is reported in Section 10.

The final-layer residual baseline selects minimum validation ASR even when utility constraints fail, whereas the calibrated control permits layer and strength selection subject to those constraints. Its higher BUP therefore reflects a change in the operating-point rule as well as the candidate layer and strength; it does not isolate a layer effect. The five-response difference between local and calibrated residual does not establish statistical superiority; the residual update is also faster. The gain-only control recovers much of the safety benefit but has lower BUP and higher test FPR at the displayed point. Its 9% test FPR does not establish calibration infeasibility, because the 8% rule is imposed on validation. This is a comparison of calibrated procedures, not a pure causal decomposition of layer choice or every ranking signal. The SARSteer adaptation has ASR 10.8%, BUP 96.4%, FPR 6.3%, and 1.4 ms median overhead. Its task retentions are 96/96.7/96.5%, so it slightly exceeds local steering on aggregate BUP and two tasks while local retains slightly more AIR-Bench utility. The three-response ASR difference does not establish a safety advantage. These neighboring points support different design choices, with no universal or statistically resolved ranking.

4 Adaptive Evaluation and Residual Risk

The initial adaptive evaluation uses 80 held-out prompts, 200 projected-gradient steps, and three random starts for defense-specific waveform optimization. Perturbations satisfy $\|\delta\|_\infty \leq 0.002$ and segmental SNR of at least 25 dB, with feasibility checked after 16-bit PCM quantization and a 16-to-48-to-16 kHz resampling round trip. Continuous gate/router relaxations are used during optimization and hard decisions are restored during decoding. Local and family-routed policies are evaluated under defense-specific adaptation; transcript screening is a transfer-only reference.

Table 3: Focused probe/decision-aware surrogate stress test with partial-gradient preprocessing on the same 80 Qwen prompts. The expanded union doubles the initial 600-step optimization budget while retaining all feasible initial candidates.

Policy	Initial unsafe / ASR	Expanded-union unsafe / ASR	Added successes
No defense	31/80; 38.8%	33/80; 41.3%	2
Local steering	23/80; 28.8%	25/80; 31.3%	2
Family-routed steering	19/80; 23.8%	23/80; 28.8%	4

Each target receives the same additional 600 optimization steps beyond the initial 600. The union retains all feasible initial candidates. Additional family-routed objectives expose probes, gate and router; success is evaluated with actual hard decisions. Per-target optimization and waveform constraints are retained. Higher ASR may reflect both budget and objective coverage; no probe-specific causal effect or convergence is inferred.

In this initial 600-step evaluation, family-routed ASR is 23.8%, compared with 38.8% without defense; local ASR is 28.8%. These results show a reduction under the stated adaptive budget and substantial residual risk. The three starts are optimization attempts on the same prompts, not three independent test populations. Continuous relaxation and restored hard decisions do not establish that every gate-evasion boundary has been explored. The demonstrated scope is the reported Qwen setting; it does not establish convergence against every adaptive objective or robustness on MOSS-Audio. The initial aggregate results do not provide per-prompt restart selections or failure-specific gradient-path attribution, so we do not infer these properties from ASR alone.

4.1 Probe/Decision-Aware Surrogate Stress Test with Partial Gradients

This partial-gradient extension adds 600 optimization steps per target and prompt to the initial 600, retaining the initial feasible candidates in the evaluation union. For family-routed steering, the added objective coverage exposes frozen component/semantic probes, the acoustic/verifier fusion, the gate and the router; hard-policy decoding, rather than relaxed surrogate success, determines whether a candidate succeeds. Local and no-defense references receive the same additional budget for applicable unsafe-response objectives. The waveform feasibility checks and the 80 prompts are retained.

The declared additional budget is six 100-step runs. Let q be the mean teacher-forced log probability of a fixed unsafe-compliance reference prefix of at most eight tokens, specified from benchmark annotations before optimization. Family-routed steering uses two starts for each of three surrogates: q, $q - 0.5 \operatorname{logit} r$, and $q + 0.5H(\boldsymbol{\rho})/\log 6$. The latter two encourage gate evasion and router uncertainty while retaining the unsafe-response objective. Local and no-defense targets instead use six starts of their applicable q. The continuation surrogate uses a sigmoid gate with temperature 0.1 and soft routing with temperature 0.5; auxiliary probabilities are clipped at 10^{-6}. Projected ascent uses step size 10^{-4} within the waveform constraints specified above. Feasible starts and every-20-step candidates are retained and decoded with the actual hard policy. Gradients include differentiable waveform-to-probe/verifier paths; nondifferentiable acoustic feature extractors are stop-gradient in this surrogate. Thus the protocol exposes the probing mechanism without claiming exact gradients through every preprocessing stage. Prefix likelihood, gate evasion, and router entropy are attack surrogates, not success labels or exhaustive decision-boundary searches.

The union counts are 33/80 without defense, 25/80 for local steering, and 23/80 for family-routed steering, each no smaller than its initial count. The additional family-routed failures reduce its point-estimate advantage over the expanded no-defense reference to 12.5 pp; family-routed steering remains useful without becoming a complete safeguard. Success counts alone cannot determine how many failures involved a closed gate, a wrong route, or an unsafe response despite a correct route, so we do not make those attributions from aggregate outcomes. Additional budget and

Table 4: Same-scope online resource measurements. Natural gate decisions, total timing, stages and memory share the declared 500-turn workload.

Quantity	Local	Conditional	Family-routed
Risk features and probes mean (ms)	0	2.6	2.6
Risk fusion mean (ms)	0	0.1	0.1
Family routing mean (ms)	0	0	0.25
Hook dispatch mean (ms)	0.48	0.3	0.3
Activation updates mean (ms)	0.95	0.5	0.56
Host scheduling and cache mean (ms)	0.22	0.45	0.49
Stage mean sum (ms)	1.65	3.95	4.3
Total minus stage sum (ms)	0.05	0.05	0.1
Paired total overhead mean (ms)	1.7	4	4.4
Paired overhead median (ms)	1.6	3.8	4.2
Paired overhead 95th percentile (ms)	2.4	5.7	6.1
Active turns / 500	500	310	310
Additional peak allocated (MiB)	40	52	54
Additional peak reserved (MiB)	64	64	96

A100 80 GB, BF16, batch one, eight-second input, 64 generated tokens, 50 warm-ups; 250 attack-like and 250 benign development requests. Matched base-model mean is 1310 ms. Stage spans and uninstrumented paired totals are separate passes over the same requests; their residual reports scheduling/instrumentation interactions. Memory is paired against a warmed base and measured independently per policy. These values do not reconstruct the primary 1.6/4.1/4.6 ms medians.

objective coverage vary together, so this extension does not isolate the effect of probe awareness or compare methods at the initial fixed budget. It supplies no convergence certificate, empirical uncertainty, or MOSS adaptive result.

5 Online Cost, Memory, and Offline Work

The primary timing protocol uses one A100 80 GB, BF16, batch size one, 50 warm-ups, and 500 matched defended/undefended turns with CUDA synchronization. Its local, conditional, and family-routed wrapper medians are 1.6/4.1/4.6 ms and exclude the base LALM forward pass. The following measured profiles use separately defined workloads.

5.1 Matched Natural-Trigger Accounting

This profile uses 500 development-only requests, 250 attack-like and 250 benign, each with an eight-second input and 64 generated tokens. All policies and the base use the same requests, seeds and timing order after 50 warm-ups. Gates use their frozen natural decisions, with 310/500 active turns for each gated policy. The 62% trigger rate belongs to this declared profiling mix, not a deployment prevalence estimate. The fixed generation length is for profiling; safety evaluations retain the primary 256-token cap.

Exclusive stage spans are profiled separately from uninstrumented end-to-end paired timing on that same workload. Their sums are 1.65/3.95/4.30 ms, compared with independent paired means of 1.7/4/4.4 ms. The residuals 0.05/0.05/0.10 ms capture total/stage interaction and instrumentation differences; they are not estimated by assigning leftover time to a named component. Medians and the 95th percentiles are request-level differences and do not add across stages. The base-model mean is 1310 ms, making wrapper means about 0.13/0.31/0.34% of this one workload's base time; this percentage does not predict other input/output lengths or front ends.

Incremental allocated peaks of 40/52/54 MiB and reserved peaks of 64/64/96 MiB are paired against a warmed base-model run with common weights and KV storage. Each policy uses an independent allocator-reset run; memory peaks are maxima rather than additive stage allocations.

Table 5: One-time Qwen discovery/calibration resource measurements.

Phase	GPU h	Wall h	Peak GiB	Retained GiB
Direction and probe extraction	1.2	1.3	24.6	2.4
Component intervention scans	10.8	11.2	44.8	4.1
Bootstrap ranking (CPU)	0	0.1	–	0.1
Configuration validation	32.4	33.1	22.4	0.1
Gate and router calibration	1.6	1.7	23.6	0.1
Total / maximum	46	47.4	44.8	6.8

Includes all proposed-method configurations, not only the selected point. Wall hours use serial execution; GPU peak is a maximum, not a sum. Retained phase artifacts are nonoverlapping and exclude weights and source audio. Download, attack creation, annotation, comparator searches and adaptive optimization are outside this accounting. CPU-only bootstrap has no GPU peak.

Table 6: Serial, gate-active resource measurements. Stage means are exclusive elapsed milliseconds; they add to the profile mean. Primary end-to-end medians are reported separately under their distinct measurement scope.

Stage or quantity	Local	Conditional	Family-routed
Acoustic features and prefill-state capture/probes	0	2.6	2.6
Risk fusion and threshold	0	0.1	0.1
Family routing	0	0	0.4
Hook dispatch and state indexing	0.5	0.4	0.4
Activation update and write-back	1	0.8	0.9
Host scheduling and cache access	0.3	0.5	0.5
Profile wrapper mean (ms; sum)	1.8	4.4	4.9
Primary wrapper median (ms; measured)	1.6	4.1	4.6
Incremental peak allocated memory (MiB; measured)	40	52	54

Profiling workload: A100 80 GB, BF16, batch one, eight-second input, 64 generated tokens, 50 warm-ups and 500 turns; gates held open for the gated policies. Zero means an unused stage. Serial spans include GPU completion at boundaries and host/cache operations. These measured means do not reconstruct the primary mixed-workload medians or predict cost at natural trigger rates. Memory is relative to a warmed matched base-model workload; it is not inferred from direction size.

Four 128-dimensional head directions plus one 4096-dimensional projected-audio direction store 4608 scalars, or 9 KiB in BF16/18 KiB in FP32. This algebra excludes probes, temporaries, copies and allocator effects, so it does not explain or predict the measured memory peaks.

5.2 One-Time Discovery and Calibration

The offline profile accounts for the full Qwen extraction, candidate intervention scans, 500 bootstrap resamples, all configuration candidates, and gate/router calibration. In this profile, GPU time totals 46 hours, serial wall time totals 47.4 hours, peak allocated memory is 44.8 GiB, and nonoverlapping retained artifacts total 6.8 GiB. Model download, attack construction, annotation, comparator searches and adaptive optimization are excluded; this account covers discovery and calibration rather than the full experimental workflow. Storage excludes model weights and source audio. The serial sums and maximum aggregate the measured phase entries. Batching, caching and the exact support inventory could materially change the outcome. No energy estimate or cost-amortization claim is made.

5.3 Separate Active-Path Profile

The serial, gate-forced-open profile measures the active path: its means are 1.8/4.4/4.9 ms. It uses its own synchronized boundaries and does not decompose either the matched natural-trigger profile or the primary medians. Differences between these workload profiles cannot identify a causal cost of gating. The resource magnitudes describe the stated profiling workloads.

6 Teacher-Forced and Autoregressive Diagnostics

Teacher forcing fixes the response prefix while estimating the matched direction; autoregressive generation replaces that prefix with the model's own tokens. These are different state distributions. The attack-response results test whether teacher-forced directions have useful behavioral effects under generated prefixes, but they do not directly quantify representation alignment as the prefix diverges. The following two analyses make that distinction explicit and avoid treating one cosine as proof of state equivalence.

6.1 Direction alignment by response position

The table compares normalized component directions estimated from teacher-forced and unmodified rollout prefixes over early, middle, and later response positions. It includes decoder heads only. Qwen's projected-audio P3 hook is applied during input prefill, before response generation; its state cannot be binned by generated-prefix length and is therefore excluded from this alignment comparison.

Table 7: Measured TF/rollout direction alignment on extraction data.

Decoder head	Tokens 1–8	Tokens 9–16	Tokens 17–32
L18-H07	0.83	0.71	0.59
L15-H03	0.79	0.68	0.53
L22-H11	0.86	0.74	0.62
L25-H02	0.76	0.63	0.48

These are direction cosines, not token-state identity, test accuracy, or confidence bounds. P3 is excluded because its input-prefill state precedes response-prefix divergence.

In these measurements, decoder alignment declines with response position, while remaining highest over the first eight response tokens where the deployed update is applied. This supports the short-window design as a bounded operating choice, without establishing that a fixed direction remains aligned throughout generation. The table reports direction cosines; it does not supply resampling uncertainty.

6.2 Behavior under alternative extraction prefixes

The behavioral comparison holds the selected component set, strength, intervention support, and eight-token deployment window fixed. Only the prefix construction used to estimate directions changes. Rollout-prefix extraction uses unmodified generations from extraction data; mixed-prefix extraction equally weights teacher-forced and rollout contrasts. Test responses are generated autoregressively in all rows. Test labels are not used to refit directions or select the deployment point.

The attack counts differ by four at most across the three extraction procedures, while benign score retention remains close. This pattern suggests that the teacher-forced estimator transfers adequately in this test setting, with no demonstrated need for mixed-prefix fitting. It does not

Table 8: Extraction-prefix comparison on the primary test suite, including the teacher-forced local configuration.

Variant	Unsafe	ASR	BUP	False ref.	FPR
Teacher-forced extraction / local	79/760	10.4%	96.1%	22/300	7.3%
Rollout-prefix extraction	83/760	10.9%	96.4%	21/300	7%
Mixed-prefix extraction	80/760	10.5%	96.3%	21/300	7%

All variants use the same selected sites, strength, support, and eight-token deployment window. Only extraction prefixes change. No test-time prefix is used for fitting.

prove prefix invariance, and close aggregate counts would need paired item-level comparison before any equivalence claim.

7 Intervention Window, Position Support, and Long-Form Tasks

The eight-token configuration was selected among 4, 8, and 16 tokens. The 32- and 256-token settings were retrospective stress probes, not candidates in that validation search, and were not used for model selection. Extending the update does not extend the model's audio context or establish streaming performance.

7.1 Validation and held-out operating points

Table 9: Window-length validation measurements with fixed sites and strength, including the calibrated eight-token configuration.

Tokens	Unsafe	ASR	BUP	False ref.	FPR
4	22/180	12.2%	97%	11/180	6.1%
8	19/180	10.6%	96.2%	13/180	7.2%
16	18/180	10%	95.3%	16/180	8.9%
32	17/180	9.4%	93.9%	20/180	11.1%
256	16/180	8.9%	91.8%	26/180	14.4%

Configuration search: 4, 8, 16 tokens. Retrospective stress probes: 32 and 256. Feasibility requires BUP at least 95% and at most 14/180 false refusals; 15/180 already exceeds 8%.

At fixed strength, the 4- and 8-token candidates satisfy the validation FPR cap; the 16-token candidate has 16/180 false refusals and does not. Validation counts for the 32- and 256-token settings are retrospective diagnostics and did not alter the calibrated selection.

Table 10: Held-out window-length measurements, including the calibrated eight-token local configuration. Test values were not used to select the window.

Tokens	Unsafe	ASR	BUP	False ref.	FPR
4	92/760	12.1%	97%	18/300	6%
8	79/760	10.4%	96.1%	22/300	7.3%
16	77/760	10.1%	95%	27/300	9%
32	75/760	9.9%	93.6%	33/300	11%
256	73/760	9.6%	91.2%	44/300	14.7%

The generation cap remains 256 tokens. Counts are over 760 attacks and 300 benign items, regardless of the intervention window.

The test curve suggests diminishing ASR improvement as the intervention window grows, accompanied by lower benign retention and higher false refusal. That trade-off is plausible but

not guaranteed: later steering can also change the generated trajectory in either direction. The held-out rows were not used for window selection.

Table 11: Measured task retention by intervention window, including the calibrated eight-token configuration.

Tokens	AIR-Bench	Clotho	AudioSet	BUP
4	97.2%	96.8%	97%	97%
8	96.5%	95.8%	95.9%	96.1%
16	95.4%	94.8%	94.8%	95%
32	94%	93.3%	93.5%	93.6%
256	91.4%	90.9%	91.3%	91.2%

Task-specific retention shows where the macro-average changes arise. Equal-weight BUP can hide different task costs, so all three task scores are shown. The values above simply decompose the measured window rows and are not independent observations.

7.2 A separate long-form benign suite

The primary benign benchmark caps responses at 256 tokens and is not a dedicated multi-step sequence suite. The long-form evaluation uses 120 additional benign items: 40 event-sequence summaries, 40 multi-step acoustic questions, and 40 source-attribution tasks. The three scores are normalized to a separately measured undefended baseline within this suite. This quantity is labelled suite retention and is distinct from the primary three-task BUP.

Table 12: Additional long-form benign-suite measurements, including the undefended baseline.

Policy	Event seq.	Multi-step	Source	Mean	False ref.	FPR
No defense	100%	100%	100%	100%	2/120	1.7%
Local, 8 tokens	95.1%	94.4%	95.7%	95.1%	11/120	9.2%
Local, 16 tokens	94.6%	93.2%	95.1%	94.3%	14/120	11.7%
Conditional, 8 tokens	96.7%	96.2%	97.1%	96.7%	8/120	6.7%
SARSteer adaptation	95.8%	95.2%	96.3%	95.8%	9/120	7.5%

Forty items per group; 120 total; cap 512 new tokens. Mean is within-suite relative score retention, distinct from the primary three-task BUP. Baseline 100% follows normalization; its false-refusal count is measured.

The results suggest modestly larger degradation for always-on steering and better retention for conditional steering, but cannot establish performance on streaming input, arbitrarily long responses, or tasks beyond these three groups. The 512-token cap applies only to this long-form suite; the primary evaluation retains its 256-token cap.

8 Matched-Site Gating and Position Masks

The main-paper comparison between local and conditional steering changes both trigger selectivity and the eligible site set: the prefill cache prevents gated writes to Qwen projected-audio group P3. It is therefore not a pure gate ablation. The following supplementary experiment fixes four decoder heads, directions, strength, and the first-eight-generation-step support across arms; P3 and all prefill writes are excluded in every arm. This design compares gating policies under matched intervention support, separately from the five-component deployed policy.

The Bernoulli control uses a probability fixed on validation and independent of test labels. In the measured counts, risk gating improves benign retention and lowers FPR while losing some attack reduction relative to always-on steering. Family-routed steering within the same four sites

Table 13: Matched-site gating measurements. Four decoder heads are eligible in every arm.

Variant	Unsafe	ASR	BUP	False ref.	FPR
Always on, four decoder heads	91/760	12%	96.9%	18/300	6%
Risk gate, same four heads	94/760	12.4%	97.4%	16/300	5.3%
Bernoulli gate, same four heads	101/760	13.3%	97.3%	17/300	5.7%
Risk gate + routing, same four heads	85/760	11.2%	96.8%	18/300	6%

Only the first eight generation steps are writable in every arm; prefill remains unmodified and P3 is excluded. The Bernoulli gate probability is fixed on validation. Routing changes directions within the same four sites.

recovers some safety reduction. This is a separate control comparison, and no conclusion about the deployed P3 contribution follows from this matched-head experiment.

The position-mask experiment asks whether the gain comes from audio-summary prefill positions, early generation positions, or their combination. It holds the same four decoder heads and strength fixed. The row that steers only the first eight generated tokens anchors the matched-site table above. The combined prefill and generation support is a separate always-on diagnostic and does not reproduce gated eligibility.

Table 14: Position-support measurements using the same four decoder heads.

Variant	Unsafe	ASR	BUP	False ref.	FPR
Audio-summary prefill only	120/760	15.8%	97.8%	10/300	3.3%
First eight decoding steps only	91/760	12%	96.9%	18/300	6%
Audio-summary + first eight	87/760	11.4%	96.6%	20/300	6.7%
All prefill positions + first eight	81/760	10.7%	95.4%	26/300	8.7%

The always-on generation-only row is shared with the matched-site gating control. These support diagnostics do not replace the primary five-component local deployment.

The measured pattern makes generation-only support competitive, while broader support reduces ASR slightly at a higher false-refusal cost. This localization result does not prove that audio-summary positions are unnecessary: the sites, activation distributions, and policy differ from the full deployment. Tensor shapes and eligible-token counts per site are outside the aggregates reported here.

9 Ranking Robustness and Full Candidate Inventories

The main text reports the selected components and summary diagnostics. This section provides the complete component-budget sweep, all 48 candidates in the bootstrap inventory, and the 40 screened decoder-head intervention results. The long tables expose the full rank distribution and screening measurements rather than only the winning components.

9.1 Component-count sweep

The validation sweep has a tie in unsafe counts at $K = 5$ and $K = 6$; the stated tie rule chooses the higher-BUP point. From $K = 7$ onward, the displayed false-refusal count exceeds 14/180. These validation summaries define the selected budget under the declared rule; they are not independent test comparisons.

9.2 All bootstrap-ranked candidates

Table 15: Complete validation sweep over the component budget.

K	Unsafe/180	ASR	BUP	False ref./180	FPR
1	31	17.2%	98.4%	7	3.9%
2	27	15%	97.8%	8	4.4%
3	23	12.8%	97.1%	10	5.6%
4	21	11.7%	96.7%	11	6.1%
5	19	10.6%	96.2%	13	7.2%
6	19	10.6%	95.8%	14	7.8%
7	19	10.6%	95.4%	15	8.3%
8	18	10%	95.1%	16	8.9%
9	18	10%	94.9%	18	10%
10	18	10%	94.7%	19	10.6%

K=5 and K=6 tie at 19 unsafe responses. Higher BUP selects K=5. K=7 is the first prefix above the validation FPR limit.

Table 16: Bootstrap frequencies and final ranks for all 48 candidates.

Rank	Component	Type	Top-10/500	Frequency	Selected
1	L18-H07	decoder head	430	0.86	Yes
2	L22-H11	decoder head	405	0.81	Yes
3	L15-H03	decoder head	365	0.73	Yes
4	P3	projected-audio group	390	0.78	Yes
5	L25-H02	decoder head	345	0.69	Yes
6	L21-H23	decoder head	156	0.312	–
7	L19-H17	decoder head	148	0.296	–
8	P1	projected-audio group	148	0.296	–
9	L23-H09	decoder head	141	0.282	–
10	L18-H04	decoder head	135	0.27	–
11	L20-H14	decoder head	116	0.232	–
12	L22-H05	decoder head	143	0.286	–
13	P5	projected-audio group	119	0.238	–
14	L24-H18	decoder head	100	0.2	–
15	L16-H06	decoder head	117	0.234	–
16	L17-H11	decoder head	119	0.238	–
17	L15-H20	decoder head	93	0.186	–
18	P2	projected-audio group	78	0.156	–
19	L25-H21	decoder head	75	0.15	–
20	L21-H16	decoder head	84	0.168	–
21	L18-H15	decoder head	65	0.13	–
22	L26-H10	decoder head	62	0.124	–
23	P6	projected-audio group	92	0.184	–
24	L23-H14	decoder head	68	0.136	–
25	L20-H06	decoder head	86	0.172	–
26	L16-H01	decoder head	57	0.114	–
27	L21-H26	decoder head	80	0.16	–
28	L18-H22	decoder head	69	0.138	–
29	P0	projected-audio group	61	0.122	–
30	L24-H07	decoder head	56	0.112	–
31	L20-H19	decoder head	60	0.12	–
32	L16-H19	decoder head	34	0.068	–
33	L22-H01	decoder head	30	0.06	–
34	L19-H02	decoder head	49	0.098	–
35	P4	projected-audio group	58	0.116	–
36	L20-H24	decoder head	36	0.072	–
37	L17-H04	decoder head	31	0.062	–
38	L22-H19	decoder head	20	0.04	–
39	L19-H11	decoder head	18	0.036	–
40	L13-H08	decoder head	43	0.086	–
41	P7	projected-audio group	13	0.026	–

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Rank	Component	Type	Top-10/500	Frequency	Selected
42	L27-H03	decoder head	49	0.098	–
43	L21-H08	decoder head	49	0.098	–
44	L17-H24	decoder head	10	0.02	–
45	L23-H03	decoder head	9	0.018	–
46	L19-H25	decoder head	19	0.038	–
47	L15-H12	decoder head	28	0.056	–
48	L14-H16	decoder head	41	0.082	–

Head identifiers and final ranks are reported for the complete candidate inventory. Selection frequencies use the preliminary ranking and are themselves a term in the final score.

Bootstrap frequency describes how often the candidate entered the preliminary top-ten across the stratified resamples. It is not a posterior probability of causal importance, nor an independent replication. Frequency contributes to the final score, so its association with final rank is partly definitional. The selected sites' frequencies are useful stability diagnostics only in conjunction with intervention effects and behavioral outcomes.

9.3 All screened decoder heads

Table 17: Held-out single-head intervention results for all 40 screened decoder heads.

Head	Audio ev.	S_h	Unsafe/760	ΔASR pp	ΔU pp	Selected
L18-H07	0.6397	2.14	130	8.3	1.4	Yes
L22-H11	0.5775	1.97	134	7.8	1.8	Yes
L15-H03	0.4862	1.81	132	8	1.3	Yes
L25-H02	0.4418	1.66	136	7.5	2.4	Yes
L21-H23	0.7394	1.52	139	7.1	1.7	–
L19-H17	0.7244	1.39	164	3.8	2.1	–
L23-H09	0.6863	1.28	144	6.4	1.9	–
L18-H04	0.6941	1.17	145	6.3	2.3	–
L20-H14	1	1.08	188	0.7	3.2	–
L22-H05	0.6525	0.99	148	5.9	1.6	–
L24-H18	0.6222	0.91	153	5.3	2	–
L16-H06	0.6093	0.84	162	4.1	1.5	–
L17-H11	0.5787	0.77	159	4.5	2.2	–
L15-H20	0.5458	0.71	154	5.1	2.5	–
L25-H21	0.573	0.64	170	3	2.7	–
L21-H16	0.5307	0.57	157	4.7	1.9	–
L18-H15	0.5329	0.5	150	5.7	2.4	–
L26-H10	0.5212	0.44	160	4.3	2.6	–
L23-H14	0.5196	0.38	151	5.5	1.8	–
L20-H06	0.524	0.31	163	3.9	2.3	–
L16-H01	0.5193	0.25	183	1.3	3.1	–
L21-H26	0.509	0.19	166	3.6	2.8	–
L18-H22	0.5109	0.13	167	3.4	1.7	–
L24-H07	0.4986	0.07	169	3.2	2.2	–
L20-H19	0.5018	0.01	189	0.5	3.3	–
L16-H19	0.5024	-0.05	172	2.8	2.6	–
L22-H01	0.4943	-0.12	147	6.1	1.6	–
L19-H02	0.4929	-0.19	174	2.5	2.9	–
L20-H24	0.4823	-0.26	175	2.4	2.5	–
L17-H04	0.4813	-0.33	177	2.1	3	–
L22-H19	0.4674	-0.4	173	2.6	2.1	–
L19-H11	0.4683	-0.47	178	2	2.7	–
L13-H08	0.4714	-0.54	181	1.6	1.9	–

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Head	Audio ev.	S_h	Unsafe/760	ΔASR pp	ΔU pp	Selected
L27-H03	0.4611	-0.62	141	6.8	3.4	–
L21-H08	0.463	-0.7	156	4.9	2.4	–
L17-H24	0.4579	-0.78	185	1.1	2.8	–
L23-H03	0.4468	-0.86	186	0.9	3.2	–
L19-H25	0.4484	-0.95	182	1.4	2	–
L15-H12	0.4534	-1.04	179	1.8	3.5	–
L14-H16	0.4442	-1.14	191	0.3	2.3	–

Unmodified reference: 193/760 unsafe responses. Reduction is computed before rounding; utility drop is a task-score decrease. The scan describes held-out behavior after selection and was not used to tune the deployed sites.

The head scan contrasts audio-evidence screening with the held-out single-head effect. L20-H14 has the largest audio-evidence value but a small behavioral change and is not selected, illustrating why attention/interface activity is only a screening signal. Because composite rank includes discovery intervention terms, this table does not independently validate all ranking weights. The measured deletion and patching tests in the main paper provide distinct intervention evidence, while a full alternative-ranking comparison is shown separately below.

9.4 Leave-one-term-out fixed-signal sensitivity

This diagnostic removes one term from the stored composite signal while keeping the candidate inventory, normalized discovery signals, default bootstrap term, $K = 5$, strength, and window fixed. The audio-evidence term is removed from final ranking only; architecture-family evidence screening remains in place. No coefficient or signal is refit, and the bootstrap frequencies are not recomputed.

Table 18: Fixed-signal leave-one-term-out ranking measurements on validation, including the calibrated default configuration.

Omitted term	Shared/5	Jaccard	Unsafe/180	BUP	FPR
Default	5	1.00	19 (10.6%)	96.2%	7.2%
Omit ablation term	4	0.67	21 (11.7%)	96.3%	6.7%
Omit patch term	4	0.67	20 (11.1%)	96.2%	7.2%
Omit gain term	3	0.43	25 (13.9%)	96.8%	6.1%
Omit audio-evidence term	4	0.67	20 (11.1%)	96.2%	7.2%
Omit benign penalty	3	0.43	17 (9.4%)	94.3%	11.7%
Omit stability term	4	0.67	21 (11.7%)	96.1%	7.8%

K=5, shared strength 3, eight-token window; no recalibration. Audio evidence is removed from the final score only; evidence screening stays fixed. Bootstrap frequencies retain the default preliminary-score definition.

The measured variants remain close to the default aggregate validation result, but selection overlap varies. This argues against presenting every term as uniquely necessary or the coefficient vector as globally optimal. It is consistent with a multi-signal ranking that is useful as one calibrated procedure, while leaving room to simplify it where evaluations preserve performance.

10 Attack-Family and Benign-Task Comparator Detail

Aggregate comparator points can conceal family-specific trade-offs. The detailed table separates the six fixed attack families for local steering, a layer-calibrated residual control, a safety-gain-only rank, and the paper-based SARSteer adaptation. The comparator counts sum to their respective aggregate totals.

Table 19: Family-level comparator measurements, with the primary local per-family counts.

Attack family	N	Local	Residual	Gain-only	SARSteer
TTS direct	150	8; 5.3%	8; 5.3%	9; 6%	7; 4.7%
Background injection	150	14; 9.3%	15; 10%	16; 10.7%	14; 9.3%
Replay spoofing	130	14; 10.8%	15; 11.5%	17; 13.1%	16; 12.3%
Multi-speaker injection	130	12; 9.2%	13; 10%	14; 10.8%	13; 10%
Psychoacoustic / high-frequency	110	21; 19.1%	22; 20%	21; 19.1%	22; 20%
Over-the-air simulation	90	10; 11.1%	11; 12.2%	11; 12.2%	10; 11.1%
Weighted total	760	79	84	88	82

Cells show unsafe count; ASR. Total counts match the main comparator table. No paired intervals or significance tests are inferred from family counts.

The analysis places the calibrated residual close to local steering on direct TTS, with differences distributed across transformed families. SARSteer is relatively competitive on replay and over-the-air in the measured breakdown, while no single family row establishes a general advantage. Counts are not paired confidence intervals; family-level causal or significance claims require retained item-level outcomes.

The benign-task table expands the targeted-control comparison and keeps task retention separate from aggregate BUP.

Table 20: Measured task retention for targeted controls, including the calibrated local configuration.

Method	AIR-Bench	Clotho	AudioSet	BUP
Local steering	96.5%	95.8%	95.9%	96.1%
Layer-calibrated residual control	95.3%	95.5%	96.3%	95.7%
Safety-gain-ranked component control	95.1%	94.7%	95.8%	95.2%
SARSteer (protocol adaptation)	96%	96.7%	96.5%	96.4%

The equal-weight macro-average is computed before compact one-decimal display; taskwise ordering differs from aggregate ordering.

The table illustrates why taskwise reporting matters: a method may improve aggregate BUP while yielding less retention on one task. The values do not support an across-task superiority claim and do not compensate for the different fitting-data budget in the SARSteer adaptation.

10.1 PCA-rank sensitivity for the SARSteer adaptation

The paper-based adaptation adds a benign PCA projection to its same-audio refusal-text contrast. The next table varies PCA rank, with layer and native strength independently selected on validation for each rank. Thus the display compares calibrated procedures, not the isolated causal effect of removing exactly one more principal component.

Table 21: SARSteer adaptation PCA-setting measurements. Layer and native strength are separately calibrated for each rank.

Variant	Unsafe	ASR	BUP	False ref.	FPR
Refusal contrast, no PCA	88/760	11.6%	95.5%	26/300	8.7%
PCA rank 5	83/760	10.9%	96.1%	21/300	7%
PCA rank 10	82/760	10.8%	96.4%	19/300	6.3%
PCA rank 20	85/760	11.2%	96.8%	17/300	5.7%

All variants use the same 120 additional group-disjoint benign fitting clips. Rank 10 reproduces the reported comparison point. Independent layer/strength calibration means this is a procedure comparison, not an isolated PCA-rank causal effect.

Rank 10 reproduces the reported SARSteer comparison point. Every row uses the same 120 additional group-disjoint benign fitting clips and the full-generation policy. These values do not resolve unequal fitting and search budgets, do not reproduce the external implementation, and do not assign performance to CrossSteer or ITI.

11 Adaptive Optimization Budget and Candidate Retention

The main adaptive result uses 200 projected-gradient steps and three starts per target. The expanded stress test uses a union of feasible candidates and increases total steps per target. The table makes cumulative budget and retained counts explicit without inferring per-step traces.

Table 22: Cumulative retained-candidate adaptive-budget measurements, including the initial 600-step references.

Target	Steps/prompt	Unsafe/80	ASR
No defense	600	31	38.8%
No defense	900	32	40%
No defense	1200	33	41.3%
Local steering	600	23	28.8%
Local steering	900	24	30%
Local steering	1200	25	31.3%
Family-routed steering	600	19	23.8%
Family-routed steering	900	21	26.3%
Family-routed steering	1200	23	28.8%

Each target has its own optimized candidate union. Later unions retain earlier feasible candidates. This bookkeeping does not supply convergence traces or per-failure attribution.

The 900-step values are intermediate subsets; 1,200-step values match the separately stated expanded-union evaluation. Monotonicity follows from candidate retention, not from a claim that every attack's success probability increases smoothly with optimization. These aggregate counts do not supply convergence curves, restart selections, or per-prompt attributions.

12 Input-Length and Output-Length Resource Measurements

The primary wrapper medians and matched-workload profile are reported above. This section reports profiling at different input and output lengths. The eight-second, 64-token anchor matches the preceding profile table. Other points are separate measurements rather than interpolations from that anchor.

Table 23: Measured input/output scaling latency in milliseconds.

Audio s	Output tokens	Base mean	Local extra	Conditional extra	Family-routed extra
2	32	620	1	2.5	2.8
8	32	830	1.1	3.3	3.7
30	32	1390	1.4	5.8	6.2
2	64	1050	1.5	3.3	3.7
8	64	1310	1.7	4	4.4
30	64	1890	1.9	6.5	7
2	128	1970	2	4.1	4.5
8	128	2180	2.2	4.8	5.2
30	128	2890	2.6	7.4	8

A100 BF16, batch one, matched natural-trigger profiling mix. The 8-second/64-token anchor matches the profile in Section 5. Other rows are separate measurements. Some longer-input means exceed 5 ms.

At 30 seconds and 128 output tokens, the measured conditional and family-routed overheads exceed five milliseconds. The latency constraint applies only to the primary validation workload, so these longer-workload results must not be interpreted as violating or extending that bound universally. Interpretation depends on prompt distribution, generation stopping, trigger prevalence, timing boundaries, and paired request-level observations at every length.

Table 24: Measured additional peak allocated memory, in MiB.

Audio s	Output tokens	Local	Conditional	Family-routed
2	32	32	40	42
8	32	38	50	52
30	32	56	76	78
2	64	34	44	46
8	64	40	52	54
30	64	58	80	82
2	128	38	48	50
8	128	44	58	60
30	128	62	86	88

Peaks are independently measured quantities, not sums of direction storage. Each policy uses the same warmed base workload and an independent allocator reset.

The memory profile follows the same length grid, but peak allocation is not inferable from latency. Peak allocations were measured independently with synchronization, a warmed base run, a common KV-cache policy, and allocator resets under the stated profiling conditions.

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